FX-Driven Debt Erosion in Local-Currency DeFi Lending: Counterfactual Evidence from MakerDAO ETH-A Draws

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Abstract

We map 130,742 MakerDAO ETH-A debt-draw events to hypothetical loans denominated in Argentine pesos (ARS) or Turkish lira (TRY), using monthly official exchange rates. The 7,105 reconstructed clean, repaid spells with exchange-rate coverage have a median observed duration of 6.95 days; under an illustrative 20% effective annual rate and stated costs, the counterfactual median net benefits over those observed durations are **−3.00**% (ARS) and **−4.14**% (TRY). At a hypothetical 12-month holding period, the corresponding medians are 2.67% and 6.59%; these are not observed-duration or realised trading outcomes. Draws from the same urn are dependent, and the largest 1% account for 72.49% and 66.85% of signed 12-month net benefits, respectively. We separate voluntary repayment from nonrecourse collateral close-out: a liquidated borrower does not repay again at the planned horizon. We do not report scenario-weighted risk-adjusted medians because terminal shocks and arbitrary weights do not identify liquidation timing or expected cash flows. The collateral exercise is a terminal-ratio diagnostic. Neither executable ARS conversion nor protocol solvency is estimated.

Keywords: Decentralized finance, foreign exchange, local-currency borrowing, MakerDAO, borrower cash flow, liquidation

1 Introduction

A borrower who owes a fixed number of local-currency units (LCU) can face a smaller USD-equivalent claim after that currency depreciates. This is a foreign-exchange accounting effect, not a measurement of consumer-price inflation. Whether it survives financing costs depends on the loan duration, position size, feasible exchange rates, and collateral outcome. A MakerDAO ETH-A draw is a DAI-denominated transaction, not evidence that anyone borrowed ARS or TRY stablecoins.

We place short reconstructed loan spells before long holding-period scenarios and distinguish voluntary repayment from nonrecourse collateral close-out. A liquidated borrower does not also repay the debt at the planned horizon. The empirical question is what observed draw sizes and timestamps imply under specified *hypothetical* local-currency debt and voluntary-repayment assumptions. The collateral analysis asks only whether a terminal mark would fall below an assumed threshold. It cannot identify an actual liquidation time, auction result, or borrower return in that branch.

The contribution is an auditable mapping from public on-chain draws and official FX rates to conditional borrower-side accounting, with robustness to repeated draws, small positions, duration, and concentration. The loss-allocation framework also explains why borrower figures do not establish a stablecoin peg or protocol solvency. Prior work on lending and liquidation motivates the distinction between

a collateral threshold and an executed sale [1, 2, 11]. Stablecoin research separates redemption and arbitrage from the value of an individual loan [3–5].

2 Related research and scope

The relevant economic mechanism is a change in the dollar value of a fixed *nominal foreign-currency liability*. It need not coincide with consumer-price inflation: an ARS or TRY price index and an ARS/USD or TRY/USD exchange rate answer different questions. A borrower who retains dollar-valued proceeds while owing a fixed number of LCU units may experience an accounting reduction in dollar debt after depreciation. Whether that accounting effect can be obtained in a market depends on an accessible conversion rate, the cost of acquiring the repayment token and the time for which the loan remains open. This is why we compare selected observed loan durations with imposed longer horizons and refrain from describing an official FX mark as an executable trade.

Research on DeFi lending describes smart-contract credit and its dependence on collateral and oracle rules [1, 2]. Empirical liquidation research illustrates why a marked collateral threshold should not be equated with an executed auction at that price [11]. A collateral sale changes ownership of the posted asset and settles as much of the debt as its proceeds permit. Treating that sale and full voluntary repayment as simultaneous costs would misstate both borrower and issuer cash flows. Our conditional ledger makes the two outcomes mutually exclusive, but the trace and FX files used here cannot identify the prices and timing needed to estimate how often either branch would occur in a hypothetical LCU vault.

Stablecoin prices additionally depend on redemption arrangements, arbitrage access and the structure of backing [3–5]. Those system-wide outcomes are not measured by an individual borrower's change in the dollar value of a liability. The research gap addressed here is therefore narrower than a test of a new protocol: we link public draw dates and amounts to official currency histories and explicit cost assumptions, then check whether the resulting descriptions survive changes in observational unit, duration and position size. The proposed balance-sheet identity specifies what additional data a solvency study would require.

3 Data and observational units

3.1 Draw events, urns, and clean spells

The public replication archive accompanying Chaleenutthawut et al. [6] contains 4,466,880 successful MakerDAO Vat traces. Using the documented debt-change calculation for `frob` [7], we retain 130,742 positive ETH-A draws of at least 1 DAI between January 2020 and July 2023, totalling 13.318 billion DAI, in 16,846 urn identifiers. The trace archive's exclusion of top-level calls limits the completeness of inferred debt histories. A draw is an event, not an independent borrower or necessarily a complete lifecycle; an urn is a vault identifier, not a verified person. The selection rules and archive checksum are in the accompanying documentation.

Table 1 Auditable sample construction. Counts reproduce `data/processed/sample_construction.csv`; successive exclusions apply after the positive ETH-A draw selection. Clean spells are a separate, restricted subset rather than all 130,742 draw events.

Stage	Count
Successful Vat call traces in public source archive	4,466,880
Decoded ETH-A debt-changing events	229,224
Positive ETH-A `frob` debt draws	137,426
Excluded before 1 January 2020	6,527
Excluded below 1 DAI after date filter	157
Final positive draw events	130,742
Distinct urn identifiers in final events	16,846
Reconstructed clean, repaid spells before FX coverage	7,947

The resulting debt-change distribution is highly skewed: the median is 4,792 DAI, the mean 101,865 DAI and the maximum 150.45 million DAI. These values are independently recoverable from the tracked processed event file. The transaction hash and urn remain attached to each event for auditing. A positive `dart` can represent a further draw from an existing vault; neither the number of rows nor the number

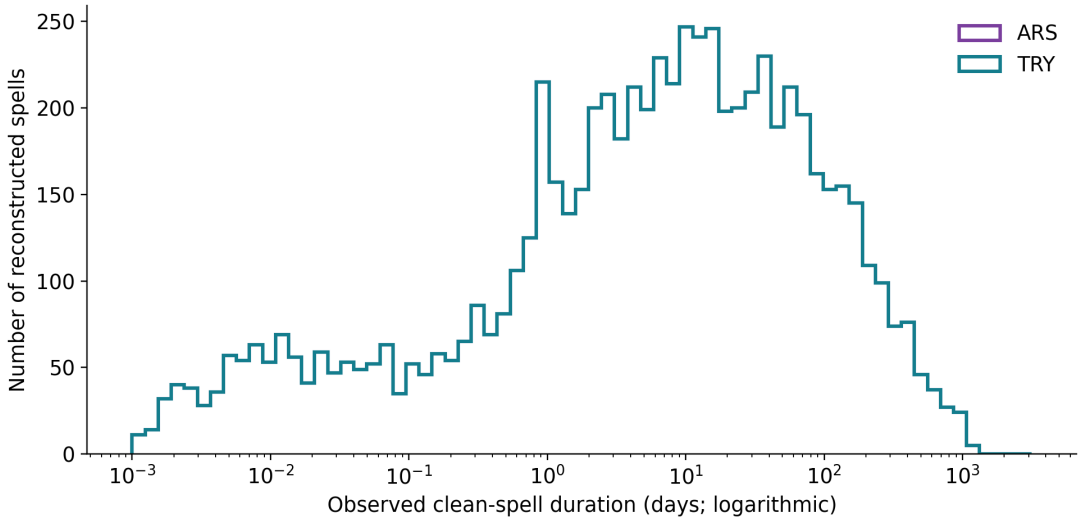


Fig. 1 Duration of 7,105 restricted clean, repaid spells with FX coverage per currency. The same selected spells are mapped to ARS and TRY; the horizontal axis is logarithmic. This subset is not a survival distribution for all draws.

of urns identifies the number of people. In particular, a draw does not establish the time at which its associated debt was ultimately repaid.

We reconstruct a restricted subset of single-draw, single-repayment, non-forked clean spells. There are 7,105 spells with usable monthly FX endpoints for each currency. Their median observed duration is 6.95 days. These are selected repaid spells and cannot estimate the duration or liquidation frequency of all vaults. Observed timestamps determine duration, while all ARS/TRY cash flows remain *counterfactual*; MakerDAO did not issue an LCU loan.

3.2 Official exchange rates

E_t denotes official LCU per USD. Archived monthly-average series `ARGCCUSMA02STM` and `CCUSMA02TRM618N` are from OECD Main Economic Indicators via FRED [9, 10]. We use the draw's calendar month and either a prespecified h-month terminal month or the clean spell's repayment month; same-month spells receive the same monthly FX rate at entry and exit. These are *official-rate counterfactuals*. An Argentine rate accessible to a user under capital controls may differ materially from this official series. No independently vetted parallel-market series is added, so none of these numbers is labelled executable. An adjacent-month timing change cannot resolve market access.

If an executable ARS rate were $E_t^{\text{exec}} = w_t E_t^{\text{official}}$ for a premium $w_t > 0$, the gross effect would be $1 - (w_0/w_h)(E_0^{\text{official}}/E_h^{\text{official}})$. An unknown change in the premium can reverse even the sign of an official-rate benefit. This identity is a sensitivity argument, not an estimated parallel-rate path.

4 Conditional borrower cash-flow model

4.1 Voluntary repayment

For a draw of P_i USD-equivalent DAI at origination, hypothetical initial LCU debt is $L_{i,0} = P_i E_{i,0}$. With an *effective annual* LCU borrowing rate r, and no intermediate payments, its USD value at voluntary closing after t years is

$$D_i(t) = P_i \frac{E_{i,0}}{E_{i,t}}(1+r)^t, \qquad G_i(t) = P_i - D_i(t)\big|_{r=0}. \tag{1}$$

The protocol fee fP_i is paid externally in USD at opening and is *not financed*. We assume a conversion friction sP_i on the opening swap of hypothetical LCU proceeds into USD and $sD_i(t)$ on acquiring LCU for voluntary repayment. The illustrative gas budget g is split equally: $g/2$ at opening and $g/2$ for voluntary closing. The incremental USD cash-accounting benefit relative to holding the same collateral without borrowing is

$$B_i^{\text{repay}}(t) = P_i - fP_i - sP_i - \tfrac{g}{2} - D_i(t) - sD_i(t) - \tfrac{g}{2}. \tag{2}$$

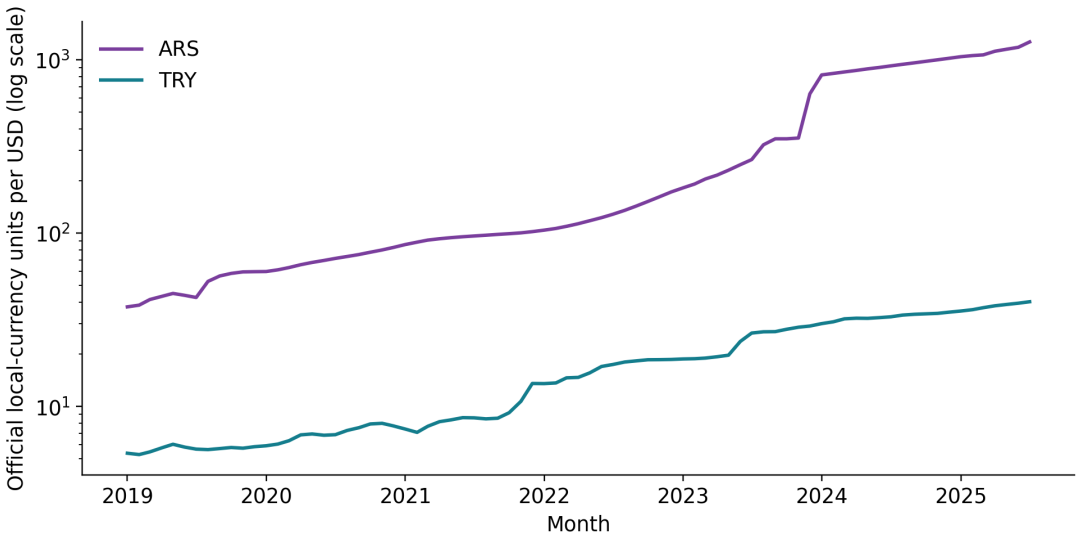


Fig. 2 Archived official monthly-average exchange rates, in local-currency units per US dollar on a logarithmic vertical axis. The series do not establish a freely executable conversion price.

Opening proceeds are assumed retained at par, with no investment return. The original collateral returns on repayment and cancels against collateral held in the no-borrowing comparator. This is neither investor profit nor welfare. Actual LCU conversions, market spread, fee funding, interest payments and taxes are unobserved. Baseline assumptions are $r = 20\%$, $f = 0.5\%$, $s = 0.3\%$ per conversion and $g = \$40$; none is an estimated price for a deployed LCU protocol.

Table 2 Base scenario conventions. All amounts and rates are assumptions of the hypothetical LCU mapping, except the specified official FX series and observed MakerDAO draw amounts and dates. Changing any one convention requires rerunning the accompanying code.

Input	Base assumption and cash-flow interpretation
Annual borrowing rate	20% effective in the base case; sensitivity rates 5, 10, 40, 80 and 120%. Nominal LCU debt accrues as $(1+r)^t$, without intermediate payments.
Protocol fee	0.5% of original USD principal, paid separately at opening and not added to financed debt.
Conversion friction	0.3% on opening principal and 0.3% on the dollar value of accrued debt at voluntary closing.
Gas budget	USD 20 at opening in either branch and USD 20 for voluntary closing; no auction gas is modelled.
Loan duration	Selected clean-spell duration, or an imposed 3-, 6-, 12- or 24-month holding scenario.
Currency conversion	Official ARS/USD or TRY/USD monthly average; no demonstrated executable ARS quote.
Collateral screen	150% terminal threshold and initial ratios of 150, 175 or 200%; no inferred liquidation probability.

The fee timing matters: charging the opening fee on discounted debt at closure would make its dollar value depend on the later FX rate and yield a different net series. We chose an externally paid opening fee and implement that same convention in every horizon and duration check. Likewise, USD 40 refers only to two specified borrower transactions; it is not an estimate of auction gas, oracle operation or all protocol transactions. Conversion friction in the liquidation branch is charged at entry only, because this stylised borrower makes no voluntary closing swap.

4.2 Mutually exclusive liquidation

If the collateral is sold at time τ rather than voluntarily repaid, its USD mark before sale is $V_{i,\tau}$. A proportional auction haircut a gives $S_{i,\tau} = (1-a)V_{i,\tau}$. Under an explicit nonrecourse accounting convention, recovered debt is $R_{i,\tau} = \min(S_{i,\tau}, D_i(\tau))$, and a penalty $\lambda D_i(\tau)$ is collected *only from*

surplus:

$$\Pi_{i,\tau} = \min\{\lambda D_i(\tau), [S_{i,\tau} - D_i(\tau)]_+\}, \tag{3}$$

$$H_{i,\tau} = [S_{i,\tau} - D_i(\tau)]_+ - \Pi_{i,\tau}, \qquad W_{i,\tau} = [D_i(\tau) - S_{i,\tau}]_+, \tag{4}$$

$$B_i^{\text{liq}}(\tau) = P_i - fP_i - sP_i - \tfrac{g}{2} - V_{i,\tau} + H_{i,\tau}. \tag{5}$$

H is cash returned to the borrower; W is the unrecovered loan claim. $V - H$ is the collateral value lost relative to holding it without borrowing. The borrower neither pays $D_i(\tau)$ again in cash nor repays $D_i(h)$ at the original horizon h. There is no voluntary exit swap or closing gas in this branch. The auction haircut and capped penalty are distinct sale effects; a different legal recourse rule would change who bears W.

For illustration, with $P = 100$, $D(\tau) = 100$, collateral worth 180, no other costs or haircut, and a 10% penalty, recovery is 100, penalty 10, residual 70, and $B^{\text{liq}} = -10$. With collateral worth 60, recovery is 60, residual and penalty are zero, and the 40 shortfall is an asset impairment. The corresponding borrower-side incremental amount is $100 - 60 = 40$ under the *assumed nonrecourse* comparator: it transfers a loss to claimholders and is not a safe trading return. Executable checks cover surplus, shortfall, haircut, cost timing and the absence of double repayment.

The available collateral shocks do not reveal τ, oracle state, auction proceeds or haircut. Equation (5) is an implemented *conditional ledger*, but this paper does not aggregate it into an expected benefit. We therefore do not report scenario-weighted risk-adjusted medians.

4.3 Terminal collateral-ratio diagnostic

For a diagnostic at a hypothetical 12-month endpoint only, initial collateral worth $c_0 P_i$ loses a stipulated fraction q and has ratio

$$\text{CR}_{i,12}^{\text{terminal}}(q, c_0) = \frac{c_0 P_i(1-q)}{D_i(1)}, \qquad I_{i,12}^{\text{terminal}} = \mathbf{1}\{\text{CR}_{i,12}^{\text{terminal}} < 1.50\}. \tag{6}$$

A breach share is the percentage of *event mappings* below an assumed 150% threshold *at month 12*. It is not the probability of ever crossing the threshold or completing an auction. The generic q values 0, 10, 25, 40 and 60% are stipulated terminal declines. ETH and BTC would produce identical results under identical percentage shocks, so we attach neither asset names nor arbitrary probabilities to the grid.

5 Empirical design and robustness

Each draw is mapped to 3-, 6-, 12- and 24-month official-rate holding *scenarios*; these are not measured MakerDAO durations. Clean spells use observed start/end timestamps and $t = \text{days}/365.25$, with monthly FX at the endpoints. We report event medians, positive shares, and $\sum_i B_i / \sum_i P_i$; the last is a *signed amount-weighted ratio*, not an average borrower's return. Means remain in machine-readable files but can be extremely negative for tiny draws facing fixed gas.

For dependence, we tabulate draws per urn, the chronologically first eligible draw from each urn, and an urn aggregate $100 \sum_{i \in u} B_i / \sum_{i \in u} P_i$. The pooled amount-weighted result is algebraically identical for all draws and urn aggregates; this is not evidence of independence. First-draw and equal-urn summaries answer different descriptive questions. To assess concentration, we rank events by *original principal size* before measuring what fraction of the full signed 12-month benefit is attributable to the largest 1, 5 and 10% of events. Month contributions use origination month. Finally, both 12-month and clean-duration mappings are rerun with minimum draw sizes of \$1, \$100, \$1,000, \$5,000 and \$10,000.

The unit changes the question. An event median weights an urn with many draws many times; selecting the first eligible draw weights each represented urn once but discards subsequent activity. The median of within-urn amount-weighted ratios also assigns one observation to each urn while allowing its own large draws to receive greater weight. None of these three summaries identifies a representative person's payoff, because control of multiple urns by one person is unverified. The event mappings also share calendar-month FX observations. Accordingly, we report descriptive differences rather than conventional independent-event standard errors.

Table 3 Observed-duration counterfactual versus fixed holding-period scenarios. All entries are percent except clean-spell counts. The observed column uses selected clean repaid durations; fixed columns map all 130,742 eligible draw events per currency. Base assumptions: effective annual rate 20%, opening fee 0.5%, conversion 0.3% each way and gas $40 round trip. Gross excludes financing and execution costs.

FX	Clean spells	Observed net	12m gross	12m net	24m net
ARS	7,105	-3.00	24.61	2.67	25.58
TRY	7,105	-4.14	25.30	6.59	37.19

Table 4 Selected clean-spell duration bands. Net values are hypothetical LCU financing benefits evaluated at observed spell lengths. N is the number of repaid spells per currency band, not people. Positive is the percentage with positive conditional net benefit.

FX	Duration (days)	*N*	Median net	Positive
ARS	(0,1]	1,891	-9.10	0.32
ARS	(1,7]	1,676	-2.54	4.89
ARS	(7,30]	1,499	-2.83	13.34
ARS	(30,90]	1,026	-1.89	24.27
ARS	> 90	1,013	-2.68	38.40
TRY	(0,1]	1,891	-9.10	0.21
TRY	(1,7]	1,676	-2.83	3.58
TRY	(7,30]	1,499	-3.66	16.68
TRY	(30,90]	1,026	-2.88	36.55
TRY	> 90	1,013	-6.05	34.06

Table 5 Twelve-month hypothetical voluntary-repayment net benefit by observational unit (percent). Each urn's aggregate amount-weighted within-urn benefit is one urn unit. Weighted net pools signed benefit divided by original principal in the selected row. Rows reuse underlying draws and are not independent estimates.

FX	Unit	*N*	Median net	Mean net	Weighted net
ARS	All draw events	130,742	2.67	-11.85	12.64
ARS	First draw per urn	16,846	0.86	-21.56	10.21
ARS	Urn aggregate	16,846	1.86	-16.48	12.64
TRY	All draw events	130,742	6.59	-6.58	27.95
TRY	First draw per urn	16,846	-4.54	-23.39	25.48
TRY	Urn aggregate	16,846	-0.43	-14.34	27.95

6 Results

6.1 Observed durations before long holding periods

The clean spell median is 6.95 days, and the hypothetical net median over *those durations* is −3.00% for ARS and −4.14% for TRY (Table 3). Only 13.03% and 14.55% of mappings are net positive. Monthly-average FX makes same-month entry and exit mechanically equal even if an intra-month market move occurred. Selection for clean repayment precludes interpreting these results as a typical return or as an estimate of all-vault duration.

For comparison, with every draw timestamp artificially held open for 12 months, gross median FX effects are 24.61% and 25.30%, while voluntary-repayment net medians are 2.67% and 6.59%. Holding for 24 months yields net medians of 25.58% and 37.19%. This chiefly contrasts *assumed horizons* with selected short spells; no typical borrower is shown to have captured the longer effect. The complete borrowing-rate grid is in `results/rate_sensitivity.csv`.

6.2 Dependence among draws

Of 16,846 urns, 6,221 (36.93%) have one eligible draw, whereas 2,713 (16.10%) have more than ten and account for 70.29% of all draw events. The complete counts and band totals are in `results/urn_draw_counts.csv` and `results/urn_draw_distribution.csv`. Treating 130,742 events as independent borrowers would give misleading precision. The TRY event median is 6.59%, but the first-draw-per-urn median is −4.54% and urn-aggregate median is −0.43% (Table 5). ARS counterparts are 2.67%, 0.86% and 1.86%. Dependence and sample composition materially affect the interpretation, and urns need not represent distinct people. The contrast is substantively larger than a rounding change:

Table 6 Effect of minimum draw size on hypothetical net benefits, under the same cost assumptions as Table 3. Observed rows refer only to selected clean, repaid spell durations. Weighted net is the sum of signed USD benefits divided by principal within each row.

FX	Sample	Min. USD	N	Median net	Weighted net
ARS	12m	1	130,742	2.67	12.64
ARS	12m	100	120,482	3.05	12.64
ARS	12m	1,000	95,125	4.86	12.65
ARS	12m	5,000	65,025	3.22	12.68
ARS	12m	10,000	50,710	3.14	12.71
ARS	Observed	1	7,105	-3.00	-0.79
ARS	Observed	100	5,795	-1.94	-0.78
ARS	Observed	1,000	3,970	-1.37	-0.77
ARS	Observed	5,000	2,642	-1.28	-0.77
ARS	Observed	10,000	2,073	-1.24	-0.77
TRY	12m	1	130,742	6.59	27.95
TRY	12m	100	120,482	10.94	27.95
TRY	12m	1,000	95,125	30.14	27.98
TRY	12m	5,000	65,025	32.37	28.06
TRY	12m	10,000	50,710	33.09	28.11
TRY	Observed	1	7,105	-4.14	0.10
TRY	Observed	100	5,795	-2.27	0.11
TRY	Observed	1,000	3,970	-1.46	0.12
TRY	Observed	5,000	2,642	-1.31	0.13
TRY	Observed	10,000	2,073	-1.27	0.13

Table 7 Concentration of hypothetical 12-month net benefit among the largest draw events, ranked by original principal. Share of signed net is the top group's sum of net USD benefit divided by the full sample's signed net USD benefit. All entries are percentages, not event probabilities.

FX	Largest draws	Share of principal	Share of signed net
ARS	1%	64.52	72.49
ARS	5%	83.41	88.03
ARS	10%	90.14	93.02
TRY	1%	64.52	66.85
TRY	5%	83.41	85.37
TRY	10%	90.14	91.69

even the sign of the TRY median depends on the unit used. It does not imply that the original event computation is arithmetically wrong; instead, a few heavily used vaults can contribute many event mappings. The draw-count distribution in the replication files lets readers see exactly how many events each urn contributes and repeat the comparison under another weighting rule.

6.3 Minimum size and contribution concentration

At the $1 floor, a fixed $40 voluntary gas budget disproportionately affects tiny positions: unweighted 12-month means are −11.85% (ARS) and −6.58% (TRY), whereas at a $100 threshold they become 5.82% and 12.74%. Small-position means are not typical returns. The five thresholds and exact denominators are in Table 6. Even above $10,000, the clean-duration net medians remain negative (−1.24% and −1.27%). The largest 1% of draws represent 64.52% of original principal and 72.49% (ARS) or 66.85% (TRY) of signed 12-month net benefit. The largest 5% account for 88.03% and 85.37%. Origination timing also matters: October 2021 contributes 13.39% of full signed ARS net benefit, while May 2021 contributes 15.21% of TRY. These accounting shares do not show that very large borrowers could access uniform FX and gas prices without market impact.

6.4 Terminal marks without expected-loss claims

With a stipulated 25% terminal decline, the ratio is below the assumed threshold for 59.92% of ARS mappings and 53.41% of TRY mappings at a 175% initial ratio. This does not reveal whether any position was liquidated earlier or a marked price was realised. Figure 4 shows all initial ratios and declines. There is no scenario-weighted expected loss or risk-adjusted headline.

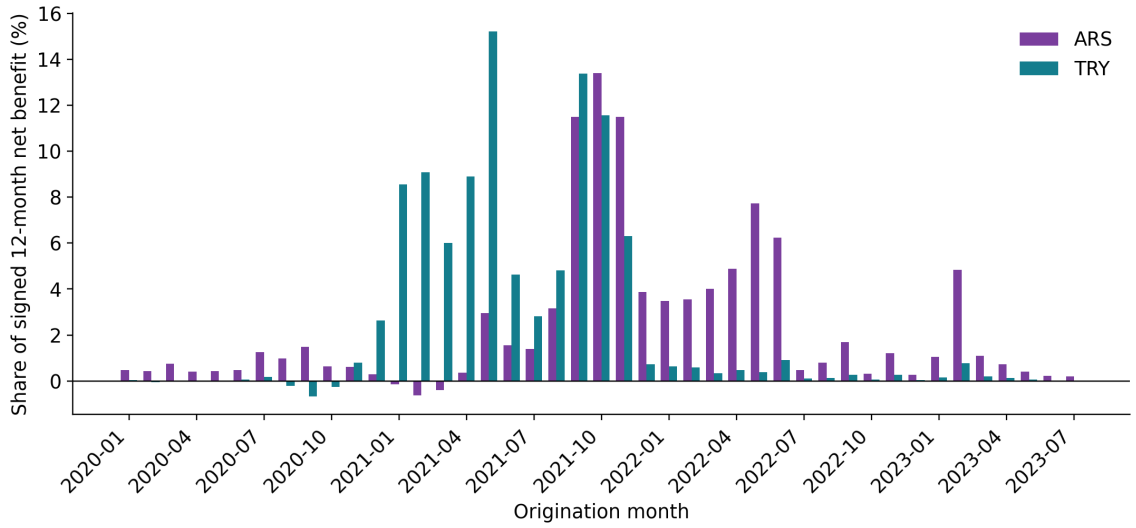


Fig. 3 Contribution by origination month to the signed amount-weighted 12-month net-benefit numerator. Shares can be negative. Complete principal weights and month results are in the accompanying CSV.

Table 8 Terminal-only screen for a 175% initial ratio, 150% threshold, 20% effective annual rate and generic stipulated collateral declines. Each entry is the percentage of 130,742 event mappings below threshold at month 12, not a liquidation probability. No estimated shock distribution, asset-specific path, auction or borrower payoff is used.

FX	Price decline	Terminal breach share
ARS	10%	0.00
ARS	25%	59.92
ARS	40%	94.48
ARS	60%	98.87
TRY	10%	3.76
TRY	25%	53.41
TRY	40%	58.29
TRY	60%	100.00

7 What the comparisons identify

The analysis contains three distinct forms of evidence. Draw timestamps, debt changes and urn identifiers come from the public MakerDAO record; the restricted clean repayment dates permit an observed spell length, subject to trace coverage and selection. The monthly official FX paths are observations of published exchange-rate series, not observations of prices paid by these borrowers. All ARS/TRY loan balances, costs and net benefits are outcomes of a stipulated conversion and repayment rule. A positive 12-month counterfactual cannot establish that an actual borrower could remain open for 12 months, obtain official-rate ARS, or retain the original collateral throughout.

The negative clean-duration medians sharpen this distinction. A same-month opening and closing month share one official monthly average, so the model records no within-month currency depreciation for that spell while still charging opening, conversion and closing costs. This mechanical property does not show that the market FX rate was constant on those days. Nor does selecting successfully repaid, single-draw, non-liquidated spells establish that all ETH-A vaults had such short lifetimes. It establishes a useful warning against presenting an imposed long horizon as an observed holding pattern.

The cost, size and concentration checks describe where the accounting result comes from. Raising a size floor changes both the fixed gas share of principal and the composition of months and urns; it does not merely remove numerical outliers. Similarly, the fraction of a *signed* aggregate attributable to the largest draws can be high even if the median event is modest. For executable gains, large trades could move the relevant market price. The current data contain neither such trading costs nor a parallel ARS quote, so we stop at a transparent conditional comparison.

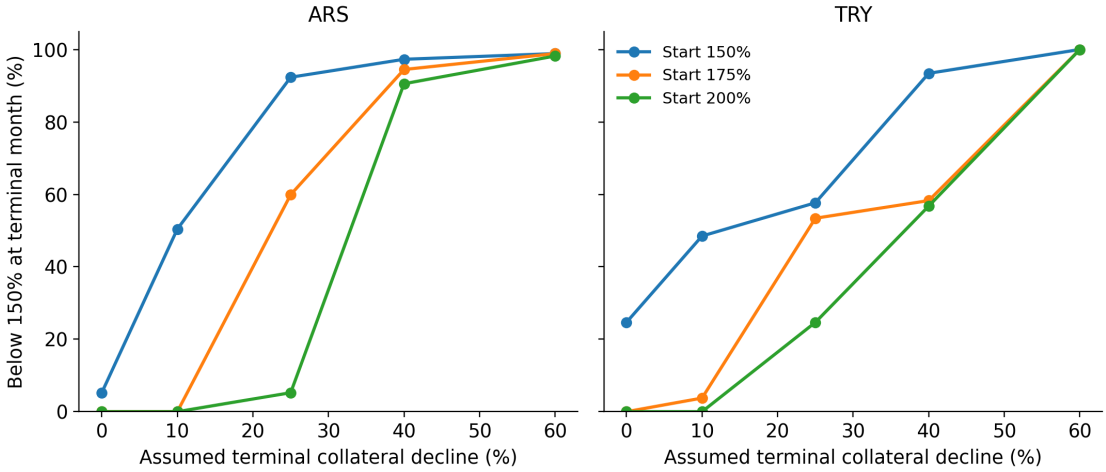


Fig. 4 Terminal ratio screen over generic deterministic collateral shocks and initial ratios. ETH and BTC are not distinguished because no asset-specific price process is estimated.

8 Illustrative balance sheet, not a solvency estimate

Borrower-posted collateral is encumbered and remains the borrower's property until enforcement; it is *not* an unrestricted issuer asset in addition to the full loan receivable. Let A_t^{loan} be the USD value of *performing* LCU loan claims; A_t^{recovery} cash received from *completed* collateral sales *after corresponding claims are removed*; and R_t unrestricted issuer reserves. Let Q_t^{redeem} be USD-valued redeemable tokens and Q_t^{other} other funding obligations. A schematic equity identity is

$$K_t = A_t^{\text{loan}} + A_t^{\text{recovery}} + R_t - Q_t^{\text{redeem}} - Q_t^{\text{other}}. \tag{7}$$

A loan undergoing liquidation is represented once: its claim is removed when cash recovery enters. Its receivable and sale proceeds cannot both be counted as free assets. At a sale, $D_i(\tau) - R_{i,\tau} = W_{i,\tau}$ writes down a loan *asset*; it is not an extra liability. A penalty collected from surplus is separate fee income. A complete issuer ledger would record any corresponding token burn, liability settlement, reserve draw and equity impact. If capital cannot absorb a shortfall, claimholders may face a depeg or loss, depending on enforceable redemption rights.

These inputs contain no observed LCU token liabilities, unrestricted reserves, auction clearing prices, redeemability terms or funding contracts. Equation (7) is a framework for later measurement, not a computed K_t or a finding about MakerDAO's actual balance sheet. Suggested rate adjustment, auction participants, diverse oracles and reserves are hypotheses for future design tests, not interventions evaluated here [5, 8, 12].

9 Limitations and interpretation

Official FX averages are not necessarily executable, especially in ARS; an unknown parallel premium has no known bias direction. Clean spells are selected and have coarse monthly FX marks. Repeated draws share urn-level risks and calendar FX shocks; no causal, confidence-interval or independent-borrower claim follows. Fixed horizons are hypothetical and cannot be inferred from clean duration. Fees, gas, conversion friction and interest rates are scenarios, not observed transaction prices for a hypothetical LCU protocol. Large positions omit market impact. The liquidation ledger assumes nonrecourse, while within-horizon collateral prices, oracles, auction discounts and settlement times are unavailable. Terminal ratios cannot identify expected cash flows. Peg stability, reserve adequacy, governance backstops, taxes and legal exchange access remain outside this dataset.

10 Conclusion

The most directly observed duration fact is that the restricted clean, repaid MakerDAO spells last a median 6.95 days. When those lengths are mapped into hypothetical local-currency debt at official monthly FX rates and illustrative costs, median net effects are negative for both ARS and TRY. Positive medians at 12 and 24 months apply to *counterfactual holding periods*; amount-weighted aggregates are

highly concentrated among large draws and particular months. The state-dependent liquidation ledger prevents double repayment, while the available collateral inputs justify only a terminal ratio sensitivity, not an expected borrower outcome. This is a transparent FX-financing accounting exercise and its limits; it does not validate a deployed protocol or its solvency.

Data and code availability

The public reproducibility package contains processed event and lifecycle files, archived official FX inputs, source query and checksum, executable borrower cash-flow model, tests, generated result tables, figures and build instructions. The repository is https://github.com/nikorokni/inflation-driven-debt-erosion-defi. The original public MakerDAO archive is linked with a checksum in the package rather than duplicated.

Declarations

Conflict of interest. No conflict of interest was reported.
Funding. No external funding was reported.
Ethics. The analysis uses public blockchain traces and published aggregate exchange rates.

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